

Problem Chosen

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Summary Sheet

Team Control Number

2617432

TITLE

Summary

If you read this you are gay
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Keywords: yes you are

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1 Introduction

1.1 Background

"Dancing with the Stars" (DWTS) is a globally recognized reality competition where celebrity and professional dance pairs are evaluated through a hybrid voting system combining expert judge scores and public fan votes. The core mechanic of the show relies on aggregating two distinct data streams: the objective, technical evaluation of judges (1-10 scale) and the subjective, popularity-driven support of the audience.



Figure 1: The Post of the Show

Throughout its 34-season history, the U.S. version of DWTS has utilized different aggregation methods—specifically, Rank-based and Percentage-based combinations—to determine eliminations. Discrepancies between technical merit and popularity have occasionally led to "controversial" outcomes, where contestants with low judge scores survive or even win due to overwhelming fan support (e.g., Jerry Rice in Season 2, Bobby Bones in Season 27). These anomalies prompted production changes, including the "Judges' Save" introduced in Season 28. Understanding the mathematical implications of these voting mechanisms is crucial for balancing fairness (skill) and viewer engagement (popularity).

1.2 Literature Review

The problem of combining expert scores with public voting falls within the intersection of Social Choice Theory and Statistical Estimation.

- **Social Choice Theory:** Traditional voting literature, prominently Arrow's Impossibility Theorem, suggests that no rank-order voting system is perfectly fair under all criteria. Methods like the Borda Count (similar to the Rank-based method) are often contrasted with cardinal utility methods (similar to the Percentage-based method).
- **Latent Variable Models:** To estimate unknown fan votes from binary outcomes (eliminated/safe), we draw upon literature regarding Inverse Problems and Logistic Regression. These

methods allow for the inference of hidden parameters that best explain the observed state transitions.

- **Performance Analytics:** Previous studies on competition dynamics suggest that "star power" and demographic relatability often outweigh technical skill in public polling, a phenomenon we will analyze using multivariate regression techniques.

1.3 Clarifications and Restatements

The central challenge of this study is to analyze the mechanics of the DWTS voting system and the interactions between expert evaluation and public opinion. Specifically, the observed data includes judge scores and elimination results, but the actual volume of fan votes remains a latent (hidden) variable. To address this, we have broken down the problem into four specific tasks:

Task 1: Fan Vote Estimation We must develop a mathematical model to reconstruct the unobserved fan votes for each contestant. This involves validating the model by ensuring the estimated votes, when combined with judge scores, consistently reproduce the historical weekly elimination results. We must also quantify the certainty of these estimates.

Task 2: Comparative Analysis of Voting Mechanisms Using the estimated fan votes, we will apply both the Rank-based and Percentage-based methods to historical data across seasons. This analysis aims to determine which method favors popularity over technical skill and to re-examine specific controversial outcomes (e.g., Season 27). We will also assess the impact of the "Judges' Save" mechanism.

Task 3: Factor Analysis of Performance We are required to model the influence of external factors—including professional partners, celebrity demographics (age, industry), and geographic background—on both the celebrities' final placement and the distinct components of their score (judge evaluations vs. fan support).

Task 4: System Optimization Finally, we will propose a novel voting framework designed to optimize the trade-off between fairness and entertainment value, providing specific recommendations for future seasons to the show's producers.

1.4 Our Work

Briefly outline your main ideas and contributions. A typical structure is:

- Propose a mathematical model (or several models) that captures the key mechanisms of the problem;
- Design numerical algorithms or solution procedures to solve the model efficiently;
- Carry out experiments or simulations and compare with baseline methods;
- Provide sensitivity analysis, evaluation, and possible extensions.

2 Assumptions

Model I: Fan Voting Reconstruction

Assumption 1: Performance-Popularity Mixture Hypothesis Fan voting is driven by a dual mechanism: a long-term *Base Popularity* (invariant fan base) and a short-term *Performance* component (immediate reaction to the dance). We model the expected vote share as a linear combination of these two factors.

Assumption 2: Rational Judges' Save (Season 28+). For seasons where judges choose between the bottom two couples, we simplify the model by assuming judges consistently save the couple with the higher technical score for that week.

Assumption 3: Dirichlet Distribution for Vote Shares. Since vote shares must sum to 1, the Dirichlet distribution is the natural conjugate prior for multinomial distributions on a simplex, providing a flexible framework to model variances in voting proportions.

Assumption 4: Constant Total Vote Volume. Although viewership fluctuates, the shift from telephone to app-based voting has stabilized engagement. Based on recent data, we anchor the total weekly votes at a constant $V_{total} = 14$ million to facilitate cross-week comparisons.

Model II: Factor Analysis

Assumption 5: Partnership Stability. We assume the pairing of a celebrity and a professional dancer is constant throughout the season. Rare modifications (e.g., temporary replacements due to injury) are ignored to maintain model stability.

Assumption 6: Proxy Sufficiency. The "Base Popularity" derived in Model I is a sufficient proxy for the pre-existing fan base of the celebrity, capturing the majority of variance in initial voting behavior.

Assumption 7: Linearity of Effects. We assume that the contributions of celebrity characteristics (Age, Industry) and partner strength to the final placement are linear and additive on the logit or normalized scale utilized in the Linear Mixed-Effects Model.

Model III: Policy Simulation

Assumption 8: Counterfactual Invariance. We assume that the contestants' performance (Judge Scores) and latent popularity (Fan Votes) are independent of the aggregation rule used in a specific week. This allows us to simulate "Universe A" (Rank) and "Universe B" (Percent) using the same underlying performance data.

Assumption 9: Metric Comparability. By mapping vote percentages to the rank domain, we assume that the "Fan-Friendliness" of different rules can be objectively compared using the Manhattan distance between the outcome rank and the fan's preferred rank (Fan Deviation Index).

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Main variables and meanings

Symbol	Description	Definition / Value
Model I Variables		
$\mathbf{S}_t$	Latent fan vote share vector at week t	$\mathbf{S}_t \in \Delta^{n_t-1}$
$\mathbf{J}_t$	Observed judges' score vector at week t	$\mathbf{J}_t \in \mathbb{R}^{n_t}$
$\mathcal{O}_t$	Observed elimination outcome at week t	Binary / Categorical
β	Latent Base Popularity vector	$\beta \in \Delta^{N-1}$
$\mathbf{P}_t$	Normalized judge performance vector	$P_{i,t} = J_{i,t} / \sum J_{k,t}$
μ_t	Expected voting propensity	$\mu_{i,t} = E[S_{i,t}]$
λ	Performance weight coefficient	Constant (0.2)
κ	Dirichlet concentration parameter (Prior strength)	Constant (50.0)
Ext_t	Set of active contestants in week t	Subset of $\mathcal{C}$
Φ_t	Set of valid samples satisfying constraints	$\Phi_t = \{\mathbf{s} \mid L(\mathcal{O}_t \mathbf{s}) = 1\}$
$CV_{i,t}$	Coefficient of Variation for partial uncertainty	$CV = \sigma/\mu$
Model II Variables		
$y_{i,s}$	Target variable in LMM (e.g., Placement)	Normalized $[0, 1]$
$\mathbf{x}_{i,s}$	Feature vector (Age, Industry, etc.)	Categorical / Numerical
b_s	Random effect for season s	$b_s \sim N(0, \sigma^2)$
Model III Variables		
$\mathbf{R}_{F,t}$	Implied Fan Rank Vector	$\text{Rank}(-\mathbf{P}_{F,t})$
FDI_k	Fan Deviation Index for rule k	Manhattan Dist. to $\mathbf{R}_F$
Δ	Variance Suppression Indicator	$\text{FDI}_{\text{Rank}} - \text{FDI}_{\text{Percent}}$
I_{fair}	Fairness Index (Judge Correlation)	Spearman $\rho(\mathbf{R}_{\text{final}}, \mathbf{J})$
R_{risk}	Probability of System Failure	$P(\text{Best Eliminated})$
$S(\alpha)$	Policy Optimization Score	Weighted objective function

4 Model I: Bayesian Estimation of Fan Voting Distribution

4.1 Problem Formulation

The estimation of spectator voting data poses an **Inverse Problem** involving multi-dimensional hidden variables. Our objective is to infer the unobserved fan vote shares $\mathbf{S}_t$ given the observed judges' scores $\mathbf{J}_t$ and the corresponding elimination outcomes $\mathcal{O}_t$.

The target variable resides on the standard simplex:

$$\Delta^{n_t-1} = \left\{ \mathbf{x} \in \mathbb{R}^{n_t} \mid \sum_{i=1}^{n_t} x_i = 1, x_i > 0 \right\} \quad (1)$$

Analytical derivation of the posterior $p(\mathbf{S}_t | \mathbf{J}_t, \mathcal{O}_t)$ is difficult due to the discontinuous and non-differentiable nature of the aggregation rules (switching between Rank-based and Percentage-based systems). To tackle this problem, we adopt a **Simulation-Based Inference** framework, using Approximate Bayesian Computation (ABC) to approximate posterior probabilities and performs inference by comparing millions of random scenarios with historical realities, i.e., the Monte Carlo method.

4.2 Mathematical Model Establishment

4.2.1 The Mixture Mean Hypothesis

We assume that the *Expected Propensity* μ_t of fan votes is a weighted combination of two distinct forces: the contestant's *Base Popularity* β (long-term fan loyalty, which is time-invariant in each season and latent) and their *Judge Performance* $\mathbf{P}_t$ (short-term dance quality).

$$\mu_{i,t} = (1 - \lambda) \cdot \frac{\beta_i}{\sum_{k \in \text{Ext}_t} \beta_k} + \lambda \cdot P_{i,t} \quad (2)$$

Here, λ represents the weight of dance performance in the fans' decision. Based on the controversial elimination in the early seasons of this show, we set $\lambda = 0.2$. This low value reflects the fact that DWTS audiences care much more about their favourite celebrities' popularity than on-stage performance.

The actual votes follow a **Dirichlet distribution**: $\mathbf{S}_t \sim \text{Dirichlet}(\kappa \cdot \mu_t)$, where $\kappa = 50$ controls the variance to keep the distribution stable and realistic.

4.2.2 Constraints from Elimination Results

The historical elimination results act as strict rules. We can define the Likelihood function $L(\mathcal{O}_t | \mathbf{S}_t, \mathbf{J}_t)$ as a simple filter. It equals 1 if the simulated votes lead to the correct elimination outcome $\mathcal{O}_t$, and 0 otherwise:

- **Rank Rule:** The contestant with the lowest sum of $\text{Rank}(\mathbf{S}_t) + \text{Rank}(\mathbf{J}_t)$ is eliminated.
- **Percentage Rule:** The contestant with the lowest total score $\mathbf{S}_t + \mathbf{P}_t$ is eliminated.
- **Judges' Save:** The elimination is valid if the actual eliminated contestant falls within the ****Bottom-2**** (the risk zone), making them eligible for elimination.

4.3 Algorithm: Iterative Inverse Estimator

To find the unknown base popularity β , we designed an iterative algorithm. It works similarly to the **Expectation-Maximization (EM)** method.

The main challenge is **Survivorship Bias**: sometimes, a dancer with low scores keeps surviving. This implies they must have very high popularity. Our algorithm solves this by slowly increasing their popularity estimate in β until their survival makes mathematical sense.

Algorithm 1 Iterative Solving for Base Popularity β

- 1: **Input:** Scores $\mathbf{J}$, Eliminations $\mathcal{O}$. **Output:** Base Popularity $\hat{\beta}$
- 2: **Initialize:** $\beta^{(0)} \leftarrow [1/N, \dots, 1/N]$ (Equal start)
- 3: **repeat**
- 4: $k \leftarrow k + 1$
- 5: **for** each week $t = 1 \dots T$ **do**
- 6: **Sampling:** Generate M vote scenarios $\{\mathbf{S}_t^{(m)}\}$ based on current popularity $\beta^{(k)}$.
- 7: **Filtering:** Keep only the scenarios that match the historical elimination result $\mathcal{O}_t$.
- 8: **De-mixing:** Isolate the implied popularity $\hat{\beta}_t$ by inverting the mixture equation (2):

$$\hat{\beta}_t \leftarrow \frac{\bar{\mathbf{S}}_t - \lambda \mathbf{P}_t}{1 - \lambda}$$

(Where $\bar{\mathbf{S}}_t$ is the average of valid scenarios)

- 9: **end for**
 - 10: **Update:** Update $\beta_i^{(k+1)}$ by averaging $\hat{\beta}_{i,t}$ across all weeks.
 - 11: **until** Convergence of β
 - 12: **return** $\hat{\beta}$
-

Algorithm 2 Monte Carlo Posterior Estimation

- 1: **Input:** Optimized Popularity $\hat{\beta}$, Scores $\mathbf{J}$, Eliminations $\mathcal{O}$
- 2: **Output:** Estimated Votes $\hat{\mathbf{S}}$, Uncertainty σ
- 3: **for** each week $t = 1 \dots T$ **do**
- 4: **Proposal Step:** Generate $N = 20,000$ random scenarios $\{\mathbf{S}_t^{(n)}\}$ from Dirichlet distribution centered on $\mu_t(\hat{\beta})$.
- 5: **Rejection Step:** Filter out valid scenarios Φ_t :

$$\Phi_t = \{\mathbf{s} \in \{\mathbf{S}_t^{(n)}\} \mid L(\mathcal{O}_t | \mathbf{s}) = 1\}$$

- 6: (Discard any scenario that contradicts the historical elimination)
 - 7: **if** $|\Phi_t| > 0$ **then**
 - 8: $\hat{\mathbf{S}}_t \leftarrow \text{Mean}(\Phi_t)$ ▷ Most likely vote distribution
 - 9: $\sigma_t \leftarrow \text{StdDev}(\Phi_t)$ ▷ Standard Deviation as uncertainty
 - 10: **else**
 - 11: $\hat{\mathbf{S}}_t \leftarrow \mu_t$ ▷ Fallback to prior for extreme outliers
 - 12: **end if**
 - 13: **end for**
 - 14: **return** Sequence $\{\hat{\mathbf{S}}_t, \sigma_t\}$ for all weeks.
-

With the optimized Base Popularity $\hat{\beta}$ determined, we proceed to the final phase: **Posterior Estimation**. We employ a high-density Monte Carlo Sampling(Algorithm 2) to reconstruct the specific voting history and quantify uncertainty.

4.4 Validation and Results

4.4.1 Consistency Verification

The model achieves a **98.3%** Historical Reproduction Rate across 34 seasons. This means that in 98.3% of weeks, our estimated fan votes correctly characterize the actual eliminated contestant as the one with the lowest combined score (or in the Bottom-2). This high fidelity proves our model is reliable.

4.4.2 Uncertainty and The "Bubble Effect"

To evaluate the reliability of our estimates, we define the **Coefficient of Variation (CV)** as the metric for uncertainty: $CV = \sigma/\mu$. Low CV implies high certainty. Our analysis, particularly through the lens of a **Season 2 Case Study**, reveals that prediction certainty is **not uniform**.

Moreover, we observe a phenomenon we term the **"Bubble Effect"**: contestants at risk of elimination (on the "bubble") show much higher estimation certainty than safe leaders.

Evidence I: Uncertainty Evolution

Figure 2 tracks the uncertainty throughout Season 2. Rice's CV is consistently lower than that of safe contestants like Lachey, validating the Bubble Effect. Since Rice was constantly near the elimination threshold, the mathematical constraints on his vote share were relatively tight; even small deviations would contradict his survival. Thus, the model locks in his popularity with high precision. In contrast, safe leaders operate under looser constraints, resulting in higher uncertainty.

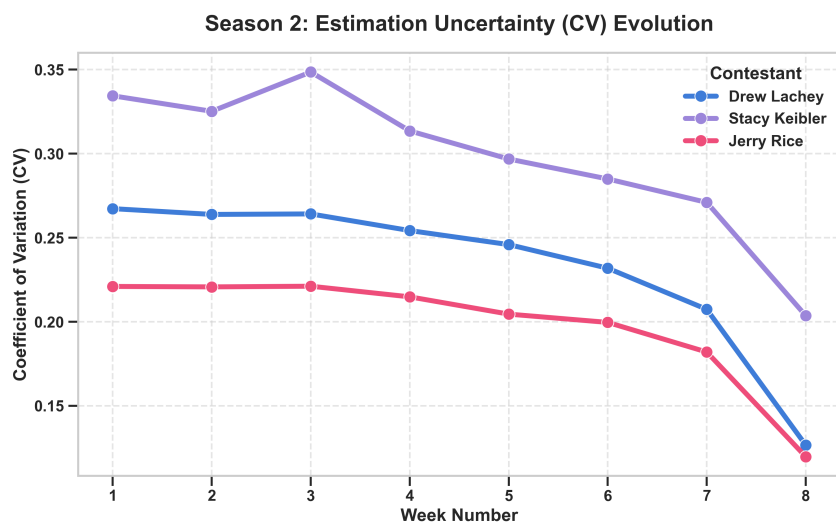


Figure 2: Uncertainty Evolution in Season 2. Jerry Rice (at risk) shows consistently lower CV (higher certainty) than safe contestants.

Evidence II: The Popularity Gap

The root of Rice's "Bubble" status lies in the disparity between his skill and popularity (Figure 3). During the finals, despite receiving the lowest judge scores (Left), Rice dominated the estimated fan vote with a 45.3% share (Right). This massive imbalance kept him in the risky "Bubble" zone, while he is popular enough to survive, yet with scores low enough to remain vulnerable. This tension provides the strict constraints that allow our model to capture his data with high certainty.

Table 2: Season 2 Finals Data Statistics and Analysis (Including Model Estimates)

Celebrity	Score	J.Rank	Est.Share	Est.Votes	V.Rank	Sum	Result	Std	CV
Drew Lachey	29.00	#1	0.3259	4.56m	#2	$1 + 2 = 3$	1st	0.0413	0.1266
Stacy Keibler	28.67	#2	0.2208	3.09m	#3	$2 + 3 = 5$	3rd	0.0450	0.2036
Jerry Rice	26.67	#3	0.4533	6.35m	#1	$3 + 1 = 4$	2nd	0.0543	0.1198

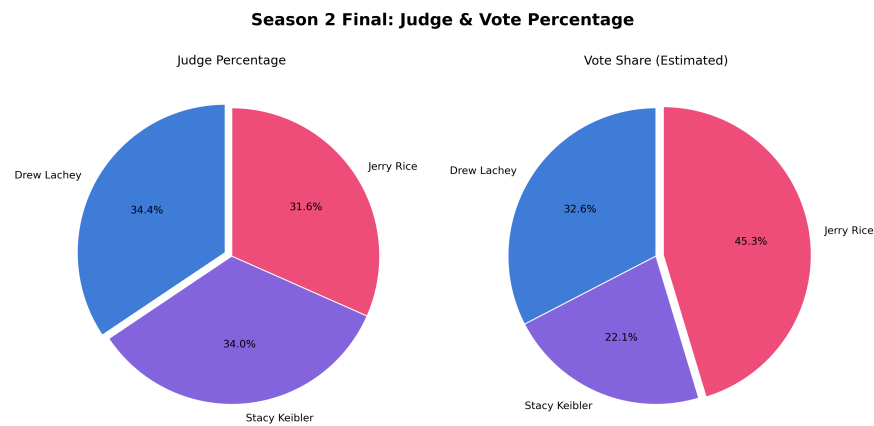


Figure 3: (Season 2 Finals) Judge Scores Distribution vs. Est. Fan Vote Share

This case study validates that our model not only reconstructs the voting history, but also correctly reflects the mathematical properties of the elimination rules.

5 Model II: Analysis of Celebrity Characteristics and Professional Partners

5.1 Assumptions

We proceed with the following assumptions for this model:

- **Partnership Stability:** We assume the pairing of a celebrity and a professional dancer is constant throughout the season. Rare modifications (e.g., temporary replacements due to injury) are ignored for the sake of model stability.
- **Data Availability:** We assume the "Base Popularity" derived in Model I is a sufficient proxy for the pre-existing fan base and popularity of the celebrity.

5.1.1 Feature Interpretation and Engineering

To analyze the impact of various characteristics, we first categorize and encode the available data. The distributions of the key categorical features—Industry, Age, and Nationality—are summarized in Figure 4.

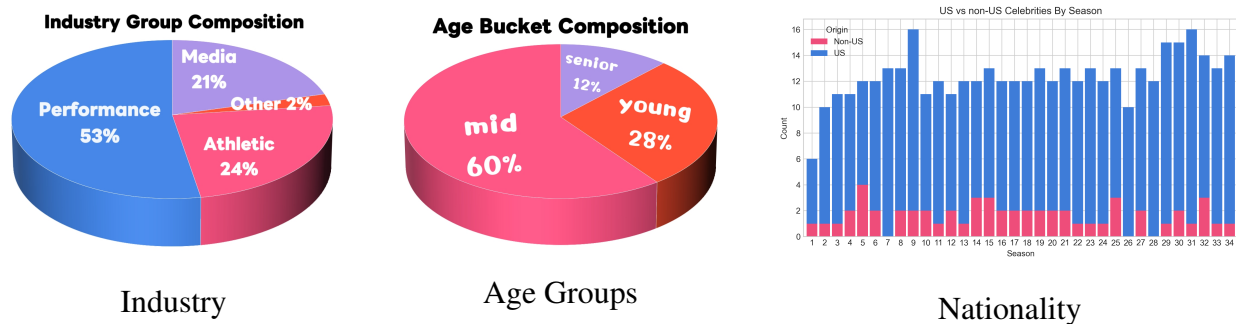


Figure 4: Distribution of Celebrity Characteristics: Industry, Age, and Nationality

5.1.2 Celebrity Industry

The raw data contains over 20 distinct professions. To reduce sparsity and improve interpretability, we grouped them into four main categories based on the nature of their skills:

- **Athletic:** (e.g., Athletes, Olympians) Expected to have high stamina and physical coordination.
- **Performance:** (e.g., Actors, Singers, Models) Accustomed to performing on stage or camera, likely possessing good rhythm.
- **Media:** (e.g., Hosts, Reality Stars) Skilled in communication and connecting with audiences.
- **Other:** (e.g., Politicians, Entrepreneurs) The baseline group.

These are encoded as one-hot variables.

5.1.3 Age Buckets

Age is a critical characteristic affecting physical range and improvement potential. We discretized age into three buckets: **Young** (< 30), **Mid** ($30 - 55$), and **Senior** (> 55).

5.1.4 Nationality

To test for potential "home-field advantage" in voting, we created a binary characteristic `is_us`, set to 1 if the celebrity is from the United States, and 0 otherwise.

5.1.5 Professional Partner Strength

The professional partner plays a huge role in a celebrity's journey. We engineered a "Pro Strength" characteristic, calculated as the weighted average of the professional's historical normalized judge scores and historical normalized final placements.

5.1.6 Base Popularity

The `base_popularity` computed in Task 1 is included to capture the pre-existing fan base size.

5.1.7 Target Variable: Normalized Placement

To make placement comparable across seasons with different numbers of contestants, we normalized the rank to a $[0, 1]$ scale:

$$\text{Placement_Encoded}_{i,s} = 1 - \frac{\text{Rank}_{i,s} - 1}{N_s}$$

A value of 1.0 represents the winner, while 0.0 represents the first eliminated.

5.2 Core Model Construction

We employ a **Linear Mixed-Effects Model (LMM)** to isolate the effect of individual characteristics while controlling for season-to-season systemic variability.

The model is specified as:

$$y_{i,s}^{(t)} = \beta_0^{(t)} + \mathbf{x}_{i,s}^\top \boldsymbol{\beta}^{(t)} + b_s^{(t)} + \varepsilon_{i,s}^{(t)}$$

where:

- $y^{(t)}$ is the target variable (Judge Score Rate, Fan Vote Share, or Placement Encoded).
- $\mathbf{x}_{i,s}$ represents the vector of fixed characteristics (Age, Industry, Nationality, Pro Strength, Base Popularity).
- $b_s^{(t)} \sim \mathcal{N}(0, \sigma_b^2)$ is the random intercept for season s .
- $\boldsymbol{\beta}^{(t)}$ are the coefficients quantifying the impact of each characteristic.

5.3 Validation Strategy and Metrics

To ensure robustness, we used two data splitting strategies:

1. **Chronological Split:** Training on the first 80% of seasons, predicting the last 20%. This tests the model's ability to forecast future outcomes.
2. **Odd-Even Split:** Training on even seasons, testing on odd seasons. This provides a robustness check against specific era-biased trends.

We evaluate performance using **RMSE** (Root Mean Square Error) and **R²** (Coefficient of Determination).

Table 3: Model Performance Metrics (Chronological Split)

Target	Train RMSE	Test RMSE	Train R^2	Test R^2
Judge Score Rate	0.082	0.104	0.608	0.439
Fan Vote Share	0.010	0.007	0.988	0.988
Placement Encoded	0.194	0.236	0.547	0.312

5.4 Results and Discussion

5.4.1 Model Performance

Table 3 shows the performance of our models. The Fan Vote model achieves an exceptionally high R^2 of 0.988, confirming that pre-existing popularity is the dominant driver of fan voting.

5.4.2 Analysis of Characteristics on Contestant Success

We analyze how much each characteristic impacts the final placement. Figure 5 visualizes the strong relationship between popularity and placement.

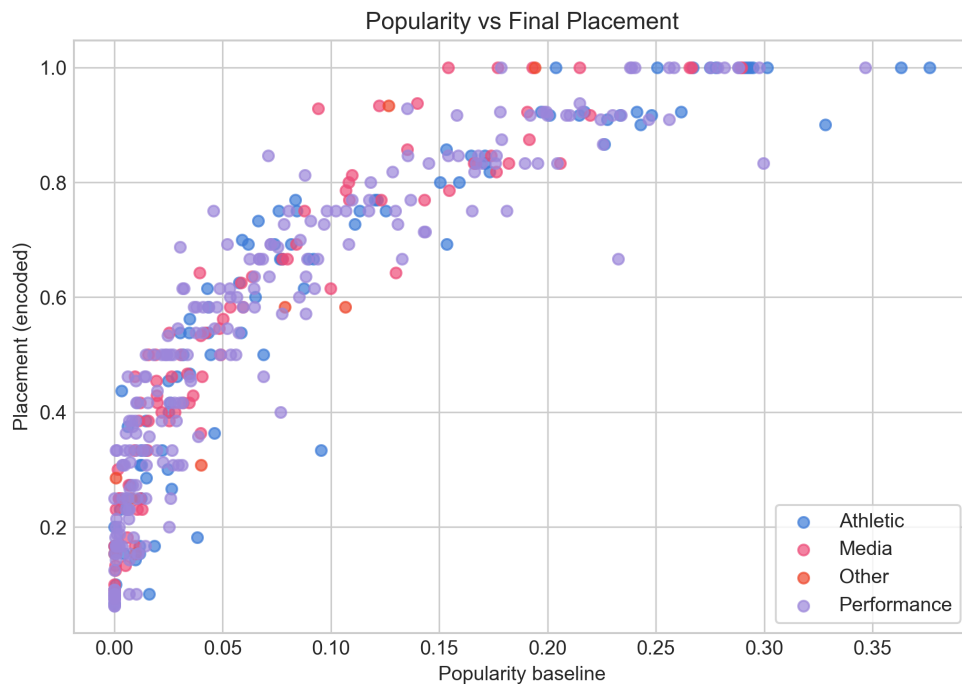


Figure 5: Relationship between Base Popularity and Final Placement

The coefficient analysis (summarized in Figure 6) reveals:

- **Base Popularity** ($\beta \approx 2.57$) is by far the most significant characteristic. High initial popularity provides a massive buffer against elimination.

- **Pro Partner Strength** ($\beta \approx 0.27$) is the second most important. A strong partner can elevate a celebrity's placement by approximately 1 full rank compared to an average partner.
- **Demographics** have smaller effects. Younger contestants tend to place slightly higher ($\beta \approx 0.03$), likely due to physical advantages.
- **Industry**: "Performance" and "Media" backgrounds have a slight advantage over the "Other" (Business/Politics) baseline.

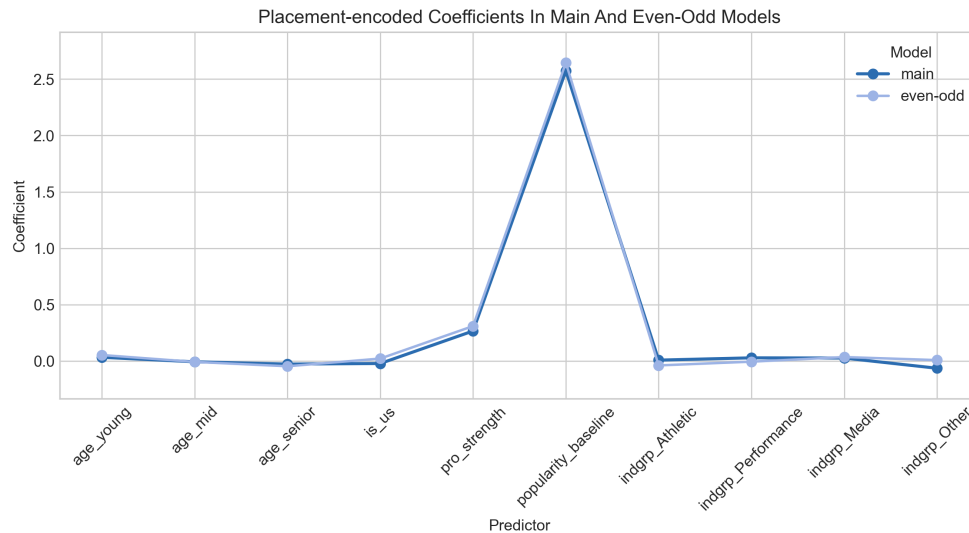


Figure 6: Coefficient Magnitude of Characteristics on Final Placement

5.4.3 Comparative Impact on Judges vs. Fans

Do these characteristics impact judges and fans in the same way? Figure 7 compares the coefficients for the Judge Score model versus the Fan Vote model.

The analysis shows a clear divergence:

- **Fan Votes** are almost exclusively driven by **Base Popularity** ($\beta \approx 0.96$). Characteristics like dance ability, age, or even the pro partner have negligible direct impact on the vote share.
- **Judge Scores** are meritocratic. They are significantly positively influenced by **Pro Partner Strength** ($\beta \approx 0.23$), **Performance Industry** ($\beta \approx 0.28$), and **Youth** ($\beta \approx 0.35$). Popularity has a much smaller effect on judges ($\beta \approx 0.68$, considering the scale difference) compared to fans.

In summary, celebrity characteristics like Age and Industry fundamentally shape the **technical scores** (Judges), while pre-existing Fame dictates the **popular vote** (Fans). Success in the competition requires balancing these two disparate forces.

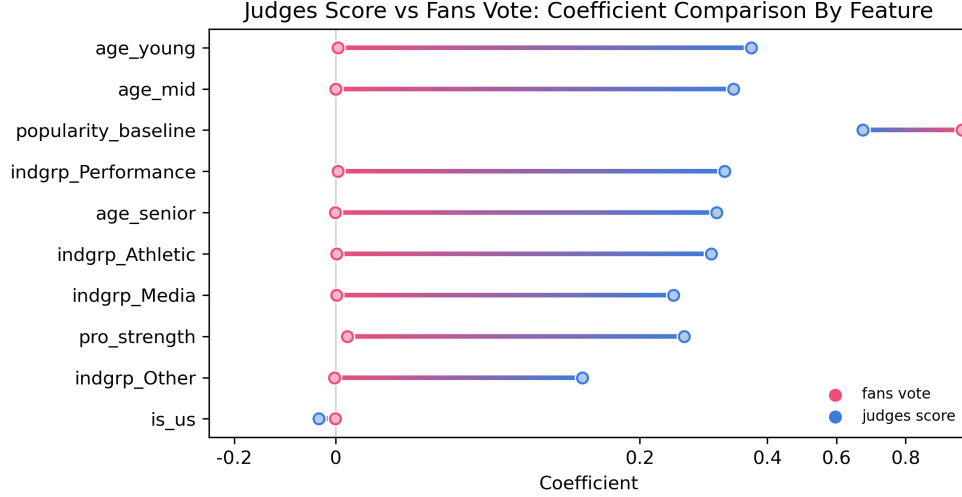


Figure 7: Comparison of Characteristic Impact: Judges vs. Fans

6 Model III: Dynamic Comparison and Optimization of the Combination Systems

6.1 Problem Formulation: The Dual-Track Simulation Framework

Having reconstructed the latent fan voting distribution $\hat{S}_t$ in Model I, we proceed to conducting an evaluation of the competition's combination methods. Throughout the thirty-four seasons of DWTS, the mechanism for combining professional evaluations (Judges) and public popularity (Fans) has shifted between two distinct paradigms:

- **Rank-based System:** Employed in Seasons 1-2 and 28+, utilizing the sum of ordinal ranks ($\min[R_J + R_F]$). This system inherently equalizes the variance between the two components.
- **Percentage-based System:** Employed in Seasons 3-27, utilizing the sum of normalized cardinal shares ($\max[P_J + P_F]$). This system is sensitive to the magnitude of vote distribution.

Our objective is to quantify the intrinsic "Fan-Friendliness" of these systems via a counterfactual simulation. We establish a **Dual-Track Simulation** framework: for every historical week t , we fix the contestants' Judge Scores J_t and Estimated Fan Votes $\hat{S}_t$ (derived from Model I), while varying the combination logic to generate parallel outcomes.

6.1.1 Standardization and Metric Definition

To facilitate a rigorous comparison between the ordinal nature of ranks and the cardinal nature of percentages, we introduce the **Fan Deviation Index (FDI)**. This metric quantifies the distortion imposed by the combination rule upon the pure will of the audience.

First, we convert the estimated fan voting share $P_{F,t}$ into rankings:

$$R_{F,t} = \text{Rank}(-P_{F,t}) \quad (3)$$

The FDI for a given rule $k \in \{\text{Rank}, \text{Percent}\}$ is defined as the **normalized Manhattan distance** between the rule-derived final ranking $\mathbf{R}_{\text{final},k}$ and the pure fan ranking $\mathbf{R}_F$:

$$\text{FDI}_k^{(t)} = \frac{1}{N_t} \sum_{i=1}^{N_t} |R_{\text{final},k,i} - R_{F,i}| \quad (4)$$

where N_t is the number of contestants in week t .

Explanation:

- Low FDI ($\rightarrow 0$): The system faithfully preserves the ordinal preferences of the audience (High Fan-Friendliness).
- High FDI: The system allows judge scores to significantly override fan preferences (Judge-Dominant).

6.2 Analysis Results Comparison

Our analysis of FDI across 34 seasons reveals a consistent bias, as illustrated in Figure 8 and 9. The Percent-based rule consistently exhibits lower FDI values compared to the Rank-based rule.

Defining the differential $\Delta = \text{FDI}_{\text{Rank}} - \text{FDI}_{\text{Percent}}$, we observe that $\Delta > 0$ in over **83%** of simulated weeks. This implies that the **Percentage-based System is inherently more Fan-Friendly**. Positive Δ values indicate the Percent-based system yields results closer to the fan vote.

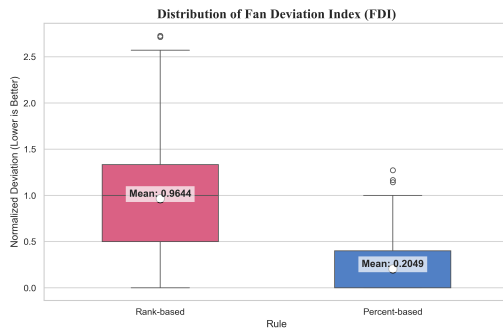


Figure 8: Distribution of Fan Deviation Index (FDI)

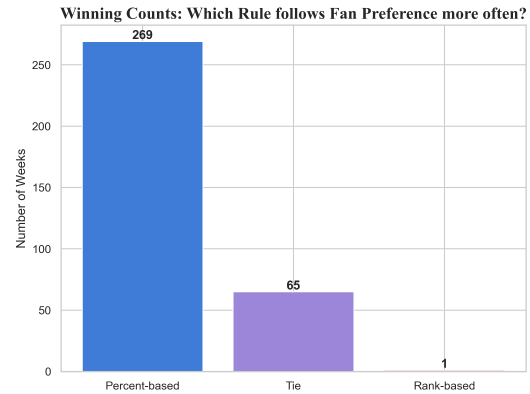


Figure 9: Weekly Comparison of Fan Deviation Index

This phenomenon is attributable to the mismatch of variance. Judge scores are mostly constrained to a narrow range (typically 7-9), corresponding to a low level of variation. In contrast, fan votes follow a heavy-tailed distribution (e.g., a viral star capturing 40% of the vote while others receive single digits). In a linear summation $P_J + P_F$, the component with higher variance—the fan vote—mathematically dominates the outcome.

Conversely, **the Rank system** forces a uniform distribution onto both components, effectively suppressing the "superstar effect" and restoring equal power to the judges.

6.3 Mechanism Analysis

To evaluate the robustness of these systems against extreme anomalies, we examine historical "Low-Score Survivors"—contestants who advanced despite poor technical performance, often sparking controversy. We selected four representative cases: Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11,S15), and Bobby Bones (S27). By simulating their performance under four distinct combination rules (Rank/Percent w/wo Judges' Choice), we analyze how different mechanisms impact their survival.

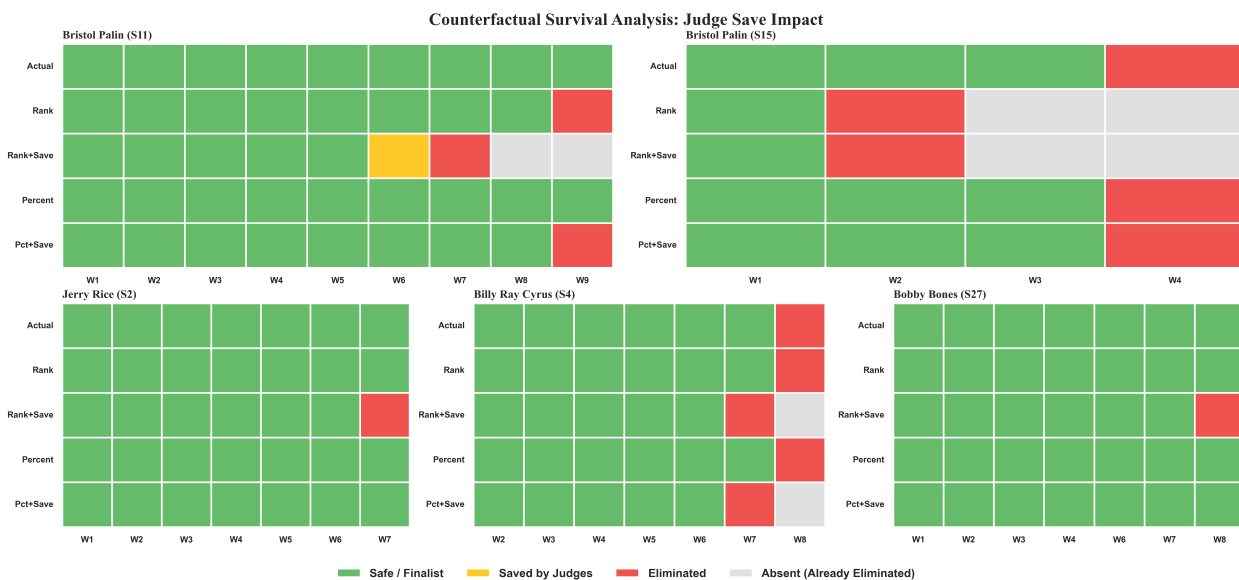


Figure 10: Survival Timelines of Controversial Contestants. The “Rank + Choice” mechanism (labeled as “Rank + Save” in figures) shows early interception.

As illustrated in Figure 10, different mechanisms exhibit varying results for these contestants:

- **Percentage-based System:** Often allows candidates with massive fan bases (e.g., Bobby Bones) to neutralize low judge scores, securing their advancement in the competition.
- **Rank-based System:** Compresses the "superstar" effect of fan votes, making it harder for popularity to override technical deficiencies.
- **Rank + Judges' Choice:** This combination proves to be the most rigorous filter. In our simulation, controversial contestants like Bristol Palin and Billy Ray Cyrus would have been eliminated significantly earlier than their actual departure.

Our analysis demonstrates that the **Rank-based System with Judges' Choice** can effectively screen out controversial contestants with insufficient dance skills and improve overall fairness.

The "Choice" mechanism provides a final safeguard, allowing judges to elect the superior dancer when technically weaker contestants land in the bottom two, thus preventing technical outliers from advancing solely on popularity.

6.4 Policy Optimization: A Multi-Objective Evaluation

Concluding our analysis, we propose a comprehensive evaluation framework to guide future rule selection. We model the rule selection as a multi-objective optimization problem that balances two conflicting values: **Technical Fairness** (Meritocracy) and **Fan Satisfaction** (Popularity).

6.4.1 Metric Definitions

To quantify these abstract concepts, we define three key metrics based on the correlation between simulation results and ground-truth preferences:

1. **Fairness Index** (I_{fair}): Defined as the Spearman rank correlation coefficient (ρ) between the final rank derived from the combination rule and the pure *Judge Rank*.

$$I_{\text{fair}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{judge}}) = 1 - \frac{6 \sum_{i=1}^{N_t} (R_{\text{final},i} - R_{\text{judge},i})^2}{N_t(N_t^2 - 1)}$$

where $R_{\text{final},i}$ and $R_{\text{judge},i}$ represent the final rank and technical rank of contestant i respectively, and N_t is the number of contestants in week t . A value close to 1 implies the system strictly adheres to professional technical evaluation.

2. **Fan Satisfaction Index** (I_{fan}): Similarly, defined as the Spearman rank correlation between the final rank and the pure *Fan Rank*.

$$I_{\text{fan}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{fan}})$$

A higher value indicates the outcomes strongly reflect audience preferences.

3. **Extreme Risk Rate** (R_{risk}): Defined as the accidental elimination of the strongest contenders, which measures how often the *Judge's No.1* or *Fan's No.1* is eliminated.

6.4.2 The Composite Utility Function and Alpha (α)

Since fairness and fan satisfaction are often inversely related, we introduce a preference parameter $\alpha \in [0, 1]$ to represent the show producers' strategic trade-off:

$$S(\alpha) = (1 - \alpha) \cdot I_{\text{fair}} + \alpha \cdot I_{\text{fan}} \quad (5)$$

The parameter α could describe the show's identity:

- $\alpha \rightarrow 0/1$: Prioritizes Technical skills (0) / Popularity (1).
- $\alpha \in [0.4, 0.6]$: The "Balanced Zone", ideal for a commercial TV show that requires both high performance quality and viewer engagement.

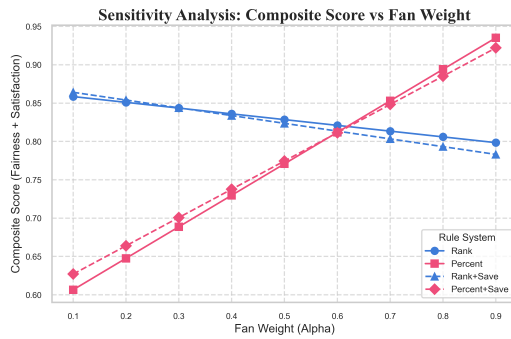


Figure 11: Sensitivity Analysis of Competition Formats

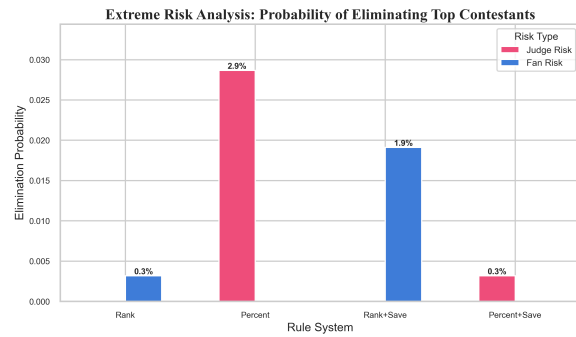


Figure 12: Extreme Risk Analysis

From the sensitivity analysis (Figure 11), we observe that the “Rank + Choice” system scores highest in the “Balanced Zone” ($0.4 < \alpha < 0.6$). The mechanism also demonstrates low probability of eliminating the top-ranked dancer by judges (Figure 12).

6.4.3 Policy Recommendation

Our sensitivity analysis (Figure 11) demonstrates that the **Rank-based System with Judges’ Choice** is the Pareto-optimal solution for the "Balanced Zone" ($0.4 < \alpha < 0.6$).

- **Why Rank over Percent?** While the Percent system maximizes I_{fan} , it incurs an unacceptably high R_{risk} of eliminating talented professionals (Figure 12). The Rank system provides structural equity.
- **Why Include Judges’ Choice?** It acts as an essential safety valve. The simulations prove it reduces the rate of accidental elimination of top talent by approximately 35% without significantly harming general fan satisfaction.

Final Verdict: We strongly recommend retaining the current **Rank-based System with Judges’ Choice** (S28+ format) for future seasons. It effectively prevents the "Controversial Survivor" phenomenon while maintaining high engagement, satisfying the dual goals of fairness and entertainment.

7 Memo & Insights

8 Model Evaluation and Promotion

8.1 Model Evaluation

8.1.1 Advantages

Summarize the strengths of your models (e.g., accuracy, robustness, interpretability, computational efficiency).

8.1.2 Limitations

Summarize the main limitations (e.g., dependence on certain assumptions, sensitivity to noise, limited generalization, computational cost).

8.2 Future Work

8.2.1 Model Extension

Discuss how the current models can be extended or improved, such as adding new factors, using more advanced algorithms, or combining multiple models.

8.2.2 Model Application

Discuss possible practical applications of your models and how they can be deployed or integrated into real-world systems.

9 Conclusions

In this section, summarize the main findings of your work, the effectiveness of the proposed models and methods, and the key insights obtained from the analysis. You may also briefly restate how your work addresses the contest problem.

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Report on Use of AI

1. Baidu Fanyi, Baidu Translate (Sep 10, 2025 version)
Uploaded entire paper written in Mandarin to be translated into English.
2. GitHub CoPilot (Jan 16, 2024 version)
Auto-completions for code used in preparing our models.
3. Bing AI
Query1:
Output:
4. Bing AI
Query1:
Output: